Project Documentation

FitnessTrack Application

Spring 2026

Capstone II

Contributions by:

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SPRINT 1:

Initial Planning 01/12/2026:

Group consensus on a Fitness Application project that incorporates features of meal plan generation, and exercise logging.

User Stories and velocity determined:

- User Story 1: User Account Registration and Login (5)
- User Story 2: Log Daily Workouts (5)
- User Story 3: Log Daily Meals (5)
- User Story 4: Generate Personalized Meal Plans (8)
- User Story 5: View Fitness Progress Dashboard (2)

Individual Works:
All team members mentioned researching resources, technology, and platforms for project implementation.

Initial Planning 01/13/2026

Mini Outline for project determined:

- Backend creation using back4app database.
- Implementation of login page mentioned, with the use of potential SwiftUI implementation.

Individual Works:

All team members on board with drawing sketches and diagrams to showcase page traversal in app.

Initial Planning 01/14/2026

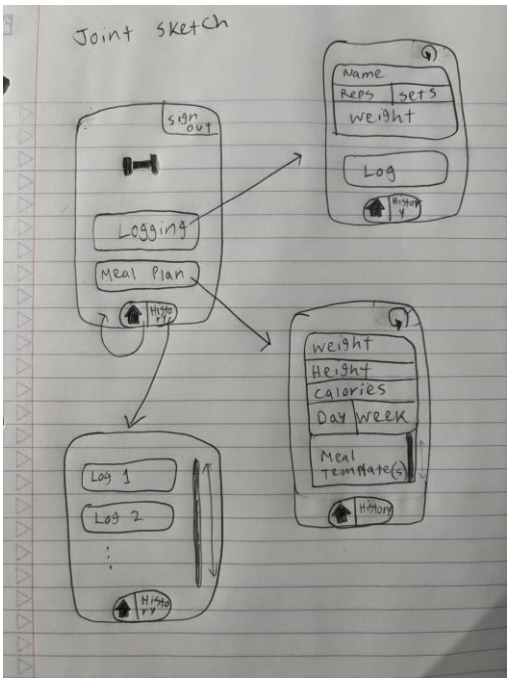
Team members have agreed to have drawn up sketches of system implementation possibilities by the following standup day.

Individual Works:

Sketches to be drawn and shown by the next standup meeting.

Initial Planning 01/15/2026

This was a joint sketch of the sketches provided by all team members. Showcasing a simple UI traversal between pages. As of right now, the best technology to do this with is SwiftUI via XCode, but we are still in the process of researching more.



Individual Works:

Keep researching further technologies to find a right fit

Initial Planning 01/16/2026

All team members have agreed to use SwiftUI via XCode implementation for the project, we were stuck on the fence between creating a web application using Java and Springboot, but ultimately chose SwiftUI for a phone application as a fitness app is more convenient in the hands of the user.

Individual Works:
We will continue to see if other platforms can be suitable in conjunction with SwiftUI.

Initial Planning 01/17/2026

Apart from the research we are doing to find suitable platforms, we have begun familiarizing ourselves with XCode and SwiftUI practices. All of us have experienced SwiftUI before, making this a good pick for the team but we need to refresh ourselves with the context of this project.

Individual works:

The team is researching and scanning documentation for ideas of implementation for the individual features designated by the user stories assigned.

Backlog Grooming Meeting Minutes 01/19/2026:

Team has designated the focus of Sprint #1 to focus and familiarize us with XCode, SwiftUI, and research relevant ideas/documentation for our features. We also designate Sprint 1 to plan and organize for future sprints.

Designated the user stories to officially commence

Individual Works:

Feature Research, and platform ideas

Initial Planning 01/20/2026

The team has continued researching ideas and documentation. We all agree that implementation of these features might be a little more challenging than we originally thought as we navigate the potential use of ParseSwift, to implement our back4app testing backend.

Individual Works:

Navigating through the benefits of ParseSwift for the back4app backend in accordance to our features.

Initial Planning 01/21/2026

The team has agreed that the use of ParseSwift is relevant enough to include, as it lets us implement our back4app application ID and our client key for secure connection. This is a crucial aspect in our development journey because it will dictate a database that all of the user stories essentially end up connecting and pulling/adding data from and to.

Individual Works: Familiarize ourselves with the use of ParseSwift.

Initial Planning 01/22/2026

Each team member has now begun properly splitting focus onto assigned user stories. Leosbell has mentioned that he began looking at API documentation for an API called Spoonacular which is essential to his meal planning generation feature. Wesley has mentioned that he began refining and simplifying his features to break them down into simpler tasks and build them up one at a time in his implementation. Sabrina has mentioned that she is still in the researching phase of the proper implementation practices for her feature.

Individual Works:

Begin applying what we have been planning to begin feature implementations.

Initial Planning 01/23/2026 - 01/25/2026

Continued our planning practices to begin feature implementation as soon as possible. All team members have mentioned they feel a lot more comfortable with the use of SwiftUI after spending time researching and understanding documentation.

Team has agreed for the retrospective that we did a good job estimating team velocity, and spending time planning in preparation for future sprints.

Individual Works:

Start applying what we have learned to begin implementations as we head into the next sprint.

SPRINT 2:

Implementations 01/26/2026

Team has agreed on the rough UI sketch above for the application layout. We have begun simple UI sketching/creation in SwiftUI before functionality implementation so we can agree on the looks of our application.

Individual Works:

Team is individually making drafts for their UI inside of XCODE before implementation of functionality.

Implementations 01/27/2026

Team is still in the process of UI development and shaping. This is the focus we have currently so we have our UI and know where to start for functionality and navigation.

Individual Works;
Continue UI-Sketching and implementation for each of our assigned respective features.

Implementations 01/28/2026

Continuing on UI-sketching, team has stated that UI-sketching is a bit difficult to do for it to have a fresh clean look, team has stated their interest in using references online as well as our sketches to combine and change around layouts.

Individual Works:

Continue UI-Sketching and implementation for each of our assigned respective features.

Implementations 01/29/2026

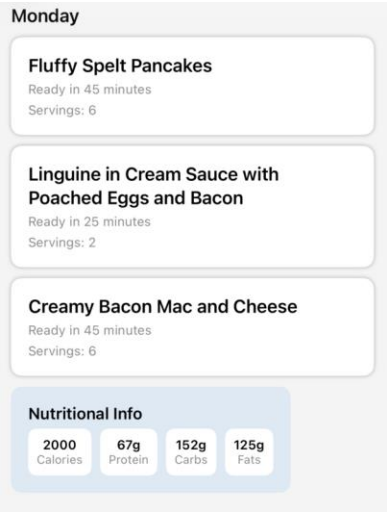
Wesley has stated he is still brainstorming for UI login page. Whether to keep it simple or decorated fully. Leosbell has stated he is currently experimenting with meal plan generation UI, to make meals appear as clean and understandable as possible. Sabrina has stated that she is experimenting with UI designs to make the most understandable UI for users.

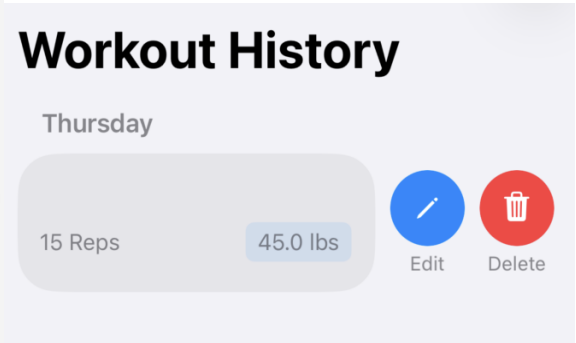
Individual Works:

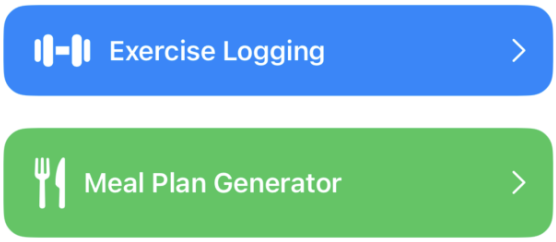
We are all still playing around with UI elements trying to come up with a clean UI that can fit all the necessary functionality needed.

Implementations 01/30/2026

We have been sharing our UI ideas with each other and have come up with some of the following designs.







Individual Works:

Continue UI work to determine best UI elements to incorporate for each of our respective features.

Implementations 02/02/2026

The team has stated satisfaction with the current UI elements incorporated into the project. We have agreed to start implementations of functionality for each feature/user story we initially created in Sprint 1.

Individual Works:

Begin implementations on each team member's own time and report progress on each standup.

Implementations 02/03/2026 - 02/06/2026

Wesley has mentioned he is UI-sketching for Fitness Dashboard, while simultaneously starting implementation of feature. Leosbell has started implementation of meal plan generation, and has shown progress on the development of API usage. Sabrina has shown progress of her feature

Individual Works;

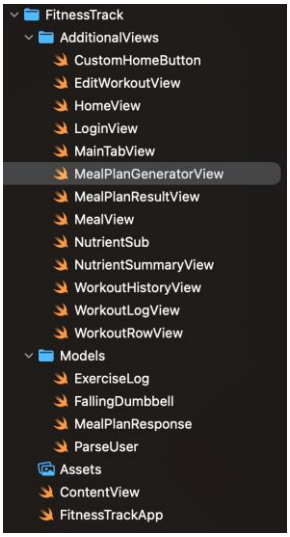
Implementation is beginning smoothly for the user stories, and team is working on each story individually on their own time.

SPRINT 3:

02/09/2026 - 02/20/2026

Progress is going smoothly, throughout this time the team has been meeting up and sharing progress. Everything so far has been going smoothly with exercise logging, simple demo of meal plan generation, etc.

The team has decided to test the project with all files combined since some features rely on other features. Example is logging history view relying on the exercise log feature. Since a feature must be implemented to be saved in the history view. To test our progress, we combined files in the following format in a single XCode simulator/build.



Separating Models, Views, and isolating the most important files of ContentView and the actual application runner “FitnessTrackApp”.

We had no hurdles, but we did need to adjust some stuff in the combination process for proper execution of the program and features to build and run.

Attached below is also our models for the back4app database.

```swift
import Foundation
import SwiftUI
import ParseSwift

struct User: ParseUser {
    var objectId: String?
    var createdAt: Date?
    var updatedAt: Date?
    var ACL: ParseACL?
    var originalData: Data?

    var username: String?
    var email: String?
    var emailVerified: Bool?
    var password: String?
    var authData: [String : [String : String]?]?
}
```

```swift
import Foundation
import SwiftUI
import ParseSwift


struct ExerciseLog: ParseObject {
    var objectId: String?
    var createdAt: Date?
    var updatedAt: Date?
    var ACL: ParseACL?
    var originalData: Data?


    var workoutName: String?
    var sets: Int?
    var reps: Int?
    var weight: Double?
    var dayOfWeek: String?
    var date: Date?
    var userId: String?
}
```

```swift
import Foundation
import ParseSwift
import SwiftUI


struct MealPlanResponse: Codable {
    let meals: [Meal]?
    let nutrients: Nutrients?
    let week: [String: DayMeals]?

    struct Meal: Codable, Identifiable {
        let id: Int
        let title: String
        let readyInMinutes: Int?
        let servings: Int?
        let sourceUrl: String?
    }

    struct Nutrients: Codable {
        let calories: Double
        let protein: Double
        let fat: Double
    }

    struct DayMeals: Codable {
        let meals: [Meal]
        let nutrients: Nutrients
    }
}
```

User Model for signup/login.

MealPlanResponse to receive JSON data from the APi successfully and also potentially save it to the database itself.

ExerciseLog serving as the model saving inputs from the user to store in the database to later display in logging history view.

Individual Works:

Build up on the combined code to implement more to our individual features.

SPRINT 4:

Implementations 03/02/2026 - 03/04/2026

Leosbell has encountered an error in JSON data mishandling for the meal plan generation feature. He has stated he will be debugging the code and then presenting the solution and reasoning to this document when ready

Individual Works:

Everybody has been implementing their features on their own time and continuining to present progress.

Implementations 03/05/2026 - 03/09/2026

The team has encountered a drastic error upon testing the combination of files again when feature testing and deploying. Sabrina has also aided Leosbell in debugging his feature of API integration.

Individual Works:

We are all individually attempting to address the issue correctly and promptly.

Implementations 03/10/2026 - 03/11/2026

Leosbell has fixed the JSON data mishandling, it was due to a mishap in the model missing carbohydrates but the MealPlanGeneratorView trying to display carbohydrates which does not “exist” in context to the model.

Fix involved simply adding: let carbohydrates: Double

Individual Works:

Wesley, and Sabrina are still debugging their respective features for an error since both of their features semi-rely on each other.

Implementations 03/12/2026 - 03/13/2026

Sabrina has found the issue with her error involving accessing data to display when it is actually non-existing which was resolved by simply adding a guard statement to check for nil cases. Wesley's issue involved being unable to save data to the database and it involved him getting an error because of a variable that we decided not to ultimately include but was still trying to be accessed in the code.

Individual Works:

Continue implementation now that we have addressed errors.

SPRINT 5:

Implementations 03/16/2026 - 03/17/2026

Everything has been going smoothly, progress has been reported accordingly in standup meetings

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Implementations 03/18/2026 - 03/19/2026

Leosbell has encountered an error in code, however it was due to Spoonacular API closing its API because of "infrastructure updates" according to his print statement when debugging.

Individual Works:

Continue working on implementations, Leosbell is awaiting the return of Spoonacular.

Print Statement Received:

API Response: <html><head> <title>Cooking a huge steak.</title><meta content="text/html; charset=iso-8859-1" http-equiv="content-type" />
<style>body{background-color:#00843c;font-size:50px;color:#fff;padding:50px;margin:0;font-family:Arial,sans-serif}</style></head>
<body> spoonacular will be back soon. We are currently upgrading our infrastructure. Thanks for spooning by! <div class="cf-error-details cf-error-

502"> <h1>Bad gateway</h1> The web server reported a bad gateway error.
<ul> <li>Ray ID: RAYIDNUMBER</li> <li>Your IP address: IPADDRESS</li>
<li>Error reference number: NUMBER</li> <li>Cloudflare Location: Atlanta</li> </ul>
 ::ALWAYS_ONLINE_NO_COPY_BOX::
</body></html>

RayID, IP address, and error reference number removed for security purposes.

Implementations 03/25/2026 - 03/26/2026

Combination testing of features results in another error for all team members. We are currently investigating the situation to address it properly.

Individual Works:

Readdress the errors received and come up with solutions that will aid the entire team to fix.

Implementations 03/27/2026

Spoonacular API's return allows for Leosbell to address and correct his error. His feature for meal plan generation is working accordingly. Sabrina and Wesley are still addressing their bugs

Individual Works:
Sabrina and Wesley are working together on their own time to address the errors in the code that affect both their features. Leosbell is implementing his own and also helping them wherever possible.

SPRINT 6:

Implementations 03/30/2026

Sabrina has fixed her error caused by a miscommunication from her feature to Wesley's feature. Wesley is still addressing an error in his own code. Leosbell has been making steady progress on his features, almost ready to demo.

Individual Works:

We are all aiding Wesley in his error for this final sprint, so we can begin polishing and updating stuff that may need it.

Implementations 03/31/2026 - 04/01/2026

Wesley made another error in code that affected Sabrina’s code, Leosbell is aiding them both to fix these errors to begin the polishing stage.

Individual Works:

Debugging these errors to begin wrapping up and readying up for the presentation.

Implementations 04/02/2026 - 04/05/2026

Wesley has fixed his bug, but his fix broke Sabrina’s code; we are currently working on a fix that can situate both features. There is an issue with how both features seemingly pull data and access it from back4app that needs to be addressed.

Individual Works:
Researching and scanning documentation to better fix features, so both do not clash.

Implementations 04/06/2026

Leosbell encountered a little error, but it is the same as last time, where Spoonacular has returned a print statement indicating an infrastructure update. Still investigating the root cause of the clashing code between Wesley and Sabrina so we can address it and fix Sabrina’s feature.

Individual Works:

Continue scanning documentation for next meetups to try and come up with error solutions.

Implementations 04/07/2026 - 04/08/2026

A possible fix has resulted in errors in the code, we had to revert back to a previous version to try and readdress the issue. Leosbell discovered an issue with how Sabrina’s code has been accessing data from the back4app backend, and thinks he might be able to find a fix.

Individual Works:

Leosbell states he will investigate the data handling portion of Sabrina’s code to finally find an ultimate fix.

Implementations 04/09/2026

Features have all been fixed and no longer clash. Clashing was a result of an accidental overwrite of data that would somehow wipe data and end up becoming nil. We have entered the polishing stage where we will review our code and improve it wherever we can while maintaining our working features.

Individual Works:

The Polishing stage has begun, we are actively monitoring, testing, and adjusting to make the application run as smoothly as possible. Preparing for the presentation.